

UNIT  
1

# DATA ANALYSIS

## SKILLS

- Reading tables and graphs
- Reading histograms and scatter plots
- Finding mean, median, and mode
- Calculating weighted averages
- Reading box plots
- Analyzing distribution
- Seeing trends and making predictions
- Analyzing correlation
- Identifying a representative sample
- Using more than one data source



**THE TERM DATA** describes a collection of facts. When the facts are numerical, you can use mathematics to analyze and draw conclusions from the data.

Data is easier to analyze when it is organized. A graph is a visual representation of data that organizes the information and provides a quick overview of the main points. In this unit, you'll study several of the most commonly used types of graphs.

Data can also be easier to analyze when it is summarized. Typical values represent what is normal, or average, in a set of data. You'll learn three types of typical values and how to use them to draw conclusions.

You can also use patterns in the data to draw conclusions. In this unit, you'll learn to recognize patterns and make predictions based on those patterns.

normal in statistics  
has a special  
meaning.

The term is:

"MEASURES OF CENTER"  
describe where the center  
of our data is.

numeric  
data  
only

any  
kind  
of data

- ① Mean - average
- ② Median - the middle value of sorted data
- ③ Mode - the most frequently occurring datum (singular)  
data (plural)

I Q  
intelligence  
quotient  
average  
math  
genius  
few people  
mentally  
retarded

compare graph  
with summary  
statistics

$$\mu = 100$$

$$\sigma = 15$$

$$21290 < 55 \text{ or } > 145$$

$$1690 < 85 \text{ or } > 115$$

$$6490 \text{ to } 85 \text{ to } 115$$

- to analyze data
- what does the data tell us?

## USING DATA ANALYSIS

You use data analysis when you:

- ✓ Read a table to find out how much medicine to give a child
- ✓ Analyze sports statistics to pick the winner of a game or event
- ✓ Interpret a graph in a magazine or newspaper
- ✓ Decide whether to believe the information in a sales pitch
- ✓ Read a bus or train schedule

Read the following items.

1. Look through a newspaper or a magazine and count how many graphs you find. Which graph is easiest to read? Why?
2. Have you ever calculated your own grade in a class or your grade point average (GPA)? How do you determine what your average should be?
3. Think of some things that increase when the temperature increases. For example, when the temperature increases, the number of people at a water park increases. What things decrease when it gets hot out?

Rokhsar Pamiry 10:32AM

The number of people wearing heavy clothes

4. Do you trust the graphs you see in newspapers or magazines or on the Internet? What makes a graph trustworthy? What makes it misleading?

Summer

heat increases:

swimming  
eat ice-cream  
we run fans

winter

heat decreases:

we wear  
heavy  
clothes  
we shovel  
snow

a special statistics term

usual: means 95% or more of data

### MATH TALK

In real life, people are not usually interested in an entire set of data. They want to know what's normal, usual, or typical. For example, if someone asks you about how much a large pizza costs, they probably don't want to know the price of a large pizza at every restaurant in the city. They just want to know that a large pizza normally costs about \$15. They want to know the typical value.

measures of center

With a partner, make a list of situations where typical values are all you really need to know. Then make a list of situations where a typical value isn't enough information.

### EXAMPLES

how many days does covid last  
if I ~~am~~ get sick?

## WORKING WITH TABLES

A **table** is often used to organize and display data, especially data that is classified by categories. A table may contain both words and numbers written in labeled rows and columns.



A **row** is read from left to right. A **column** is read from top to bottom.

Use the row and column headings to locate specific data in a table. The title of the table may also have important information.

**Population of the Top 5 U.S. Cities (2000 and 2010)**  
(in millions)

| City             | 2000 Population | 2010 Population | 2010 Rank | Percent Change 2000–2010 |
|------------------|-----------------|-----------------|-----------|--------------------------|
| Chicago, IL      | 2.89            | 2.69            | 3 ✓       | –6.9% <i>decr</i>        |
| Houston, TX      | 1.95            | 2.10            | 4         | +7.5%                    |
| Los Angeles, CA  | 3.69            | 3.79            | 2 ✓       | +2.6%                    |
| New York, NY     | 8.01            | 8.17            | 1 ✓       | +2.1%                    |
| Philadelphia, PA | 1.51            | 1.52            | 5         | +0.6%                    |

Source: U.S. Census, 2000 and 2010

$$\begin{array}{l} \text{Houston} \quad \times 4 = \text{NYC} \\ 2 \text{ mil} \quad \times 4 = 8 \text{ mil} \end{array}$$

*multiplication*

### FINDING INFORMATION IN TABLES

**Example** In 2000, the population of New York City was about how many times larger than the population of Houston?

Locate the place where the column labeled “2000 Population” and the row labeled “New York, NY” intersect.

The value is 8.01. The title states that the values are in millions. The population of New York City was about 8.01 million in 2000.

Now look for the row labeled “Houston, TX.” The population of Houston was about 1.95 million in 2000.

$8.01 \div 1.95 \approx 8 \div 2 \approx 4$ . The population of New York City was about **4 times** larger than the population of Houston in 2000.

|              | 2000 Population |
|--------------|-----------------|
|              | 2.89            |
|              | 1.95            |
|              | 3.69            |
| New York, NY | 8.01            |
|              | 1.51            |

Step 4: **Interpret** For example, the population of New York City was about 4 times larger than the population of Houston in 2000.

You can also use the data in a table to draw conclusions. For example, look at the column labeled "Percent Change." Chicago was the only city with a decreasing population. You could conclude that from 2000 to 2010, the populations of the largest U.S. cities primarily increased.

## Tables: Examples #1 & #2

Wednesday, December 11, 2024 11:53 AM

Use the following table for problems 1 and 2.

Summary of Saturday Price Reductions

| Item             | Original Price     | Discount Rate | Sale Price |
|------------------|--------------------|---------------|------------|
| Portable Speaker | \$80               | \$12 15%      | \$68       |
| Earbuds          | \$18               | \$0.90 5%     | \$17.10    |
| MP3 Player       | \$26               | \$2.60 10%    | \$23.40    |
| DVD Player       | 12 ÷ 4<br>\$120    | \$30 25%      | \$90       |
| HD Television    | -50<br>+4<br>\$230 | \$46 20%      | \$184      |

1. What is the difference in price between a portable speaker and a DVD player on Saturday?

\$68

\$90

\$22

2. Is the actual discount (in dollars) on an HD television greater than or less than the actual discount on a DVD player?

\$46

\$30

HD more than DVD

Use the table for problems 3-6.

1 ton = 2000 lbs

Yard Waste Recycled by County (in tons)

| Month    | Carroll County | Logan County |
|----------|----------------|--------------|
| January  | 520            | 1,345        |
| February | 1,080          | 2,100        |
| March    | 990            | 2,980        |
| April    | 2,250          | 3,405        |

what if we interpreted the question as:

RELATIVE INCREASE?

$$\frac{\text{change}}{\text{start}} = \%$$

$$\frac{\text{change}}{\text{start}}$$

$$\text{Jan} \rightarrow \text{Feb} \quad 755 / 1345 = 56\% \text{ increase}$$

$$\text{Feb} \rightarrow \text{Mar} \quad 880 / 2100 = 42\% \text{ increase}$$

$$\text{Mar} \rightarrow \text{Apr} \quad 425 / 2980 = 14\% \text{ increase}$$

3. How much more yard waste did Logan County recycle in February than Carroll County?

$$2100 - 1080 = 1020 \text{ tons}$$

$$(2100 - 1100 + 20)$$

5. During which month did Logan County's recycling increase the most compared to the previous month?

March

880 tons  
correct: 755 tons 56% inc

4. During which month did Carroll County's recycling decrease compared to the previous month?

90 tons

6. From January to April, which county's recycling increased the most, and by how much?

Jan → April

Carroll 520 to 2250. Increase of 1730 tons. 333%

Logan 1345 to 3405. Increase of 2060 tons. 153%

NRP answer: Logan 2060 tons

The Sparrow's answer: Carroll 1730 tons  
How is it possible to say 1730 is greater than 2060?  
Relative Increase !!

# Tables: Examples #7 & #8

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11:56 AM

Use the mileage chart for problems 7 and 8.

Airline Mileage to North American Cities

| Cities → ↓    | Mexico City       | Montreal         | New York        | San Francisco     |
|---------------|-------------------|------------------|-----------------|-------------------|
| Mexico City   | <del>XXXXXX</del> | 2312             | 2085            | 1883              |
| Montreal      | 2312              | <del>XXXXX</del> | 330             | 2537              |
| New York      | 2085              | 330              | <del>XXXX</del> | 2566              |
| San Francisco | 1883              | 2537             | 2566            | <del>XXXXXX</del> |

7. How many miles will Armando travel if he flies from Mexico City to San Francisco and continues to New York?

Mex → SF 1883  
 SF → NYC 2566  


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 4449 miles

8. Explain Why are there four empty boxes on this chart?

Because it is a city distance to itself is zero.

# Quantitative & Qualitative Data

Wednesday, December 11, 2024 11:50 AM

Quantitative Data is numeric data for which math operations can be performed.

For Example:

- Age ✓
- Height
- Weight
- Length
- Volume
- Pressure

these numbers  
are NOT  
quantitative

are attributes or  
categories or  
labels

Qualitative Data is nonnumeric data for which math operations cannot be formed.

For Example:

- Identity Number E.g., 372-81-5454
- Street Address
- Movie Titles ✓
- Authors ✓
- Painters ✓
- Colors ✓
- Foods ✓
- GENDER
- CITIES
- COUNTRIES

**Notice:** Some data consists of numbers, but because we can't perform math operations on these numbers (such as your street address), those numbers are considered non-numeric.

# Bar Graphs (Qualitative Data) & Histograms (Quantitative Data)

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non-numeric

numeric

## BAR GRAPHS AND HISTOGRAMS

Graphs give a visual picture of data. In a **bar graph**, data is represented by the height (or length) of **data bars**. The bars may be drawn vertically or horizontally.

For the bar graph below, you can use the **title** (Cafeteria Orders at Newark High School) and **axis labels** to determine what values are being compared. The **horizontal axis** shows the type of meal ordered. The **vertical axis** shows the number of meals ordered during the week.

OR

=====

QUALITATIVE DATA ONLY

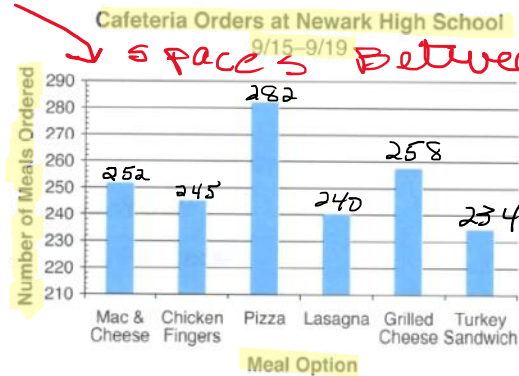
### FINDING INFORMATION ON BAR GRAPHS

**Example** About how many orders of chicken finger meals were sold at Newark High School from 9/15 to 9/19?

245

Find the bar labeled Chicken Fingers. Read the value on the vertical axis closest to the height of the bar.

You may need to estimate values on a bar graph. The bar for chicken fingers is about halfway between 240 and 250. **About 245 chicken finger meals** were sold from 9/15 to 9/19.



spaces between the bars

"about 42"

Use the bar graph above for problems 1 and 2.

1. Which meal options sold over 250 meals from 9/15 to 9/19?

Mac & Cheese  
Pizza  
Grilled Cheese

2. About how many more orders were for pizza than for lasagna?

42

# Bar Graphs: Examples

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Qualitative  
(non-numerical)

Use the double bar graph for problems 3-6.

3. About how much more did the petting zoo make in Year 2 than Year 1?

$$\begin{aligned} \$4K - \$2.5K &= \\ \$1.5K \end{aligned}$$

4. In which event did profits decrease from Year 1 to Year 2?

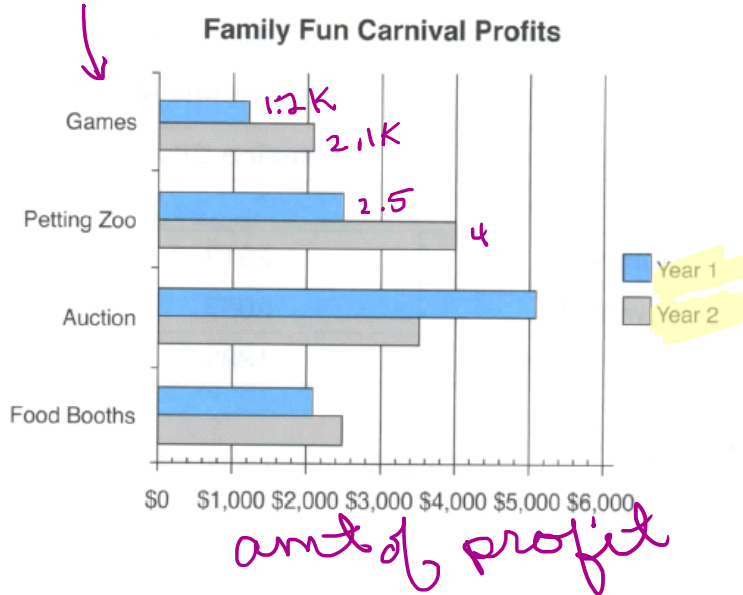
Auction

5. In Year 1, about how much more money did the auction make than the food booths?

$$\begin{aligned} \$5.1K - \$2.1K &= \\ \$3K \end{aligned}$$

6. Which event's profits increased the most from Year 1 to Year 2?

not Relative Incr  
Games? Petting Zoo



# Histograms (Quantitative Data)

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A **histogram** also uses data bars to visually show data. On a bar graph, the bars represent categories. On a histogram, the bars represent a range of numbers. You can use a histogram to draw conclusions about how frequently events occur.

Each bar on the histogram below represents an interval. For example, the first bar represents the number of cars sold with an estimated gas mileage of 5 to 9 miles per gallon. Use the vertical axis to read the value of the first bar. Two cars were sold with an estimated gas mileage between 5 and 9 miles per gallon.

## FINDING INFORMATION ON HISTOGRAMS

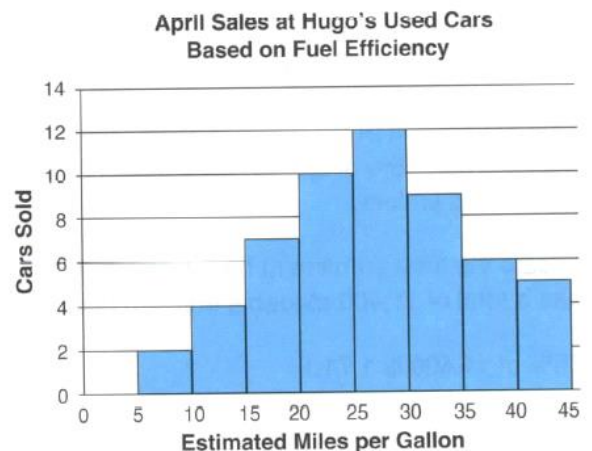
**Example** How many cars were sold in April with an estimated gas mileage of less than 15 miles per gallon?

Find 15 on the horizontal axis. The bars to the left of this marker represent cars with a mileage of less than 15 mpg.

Read the values for both bars and find the total.

|           |             |
|-----------|-------------|
| 5 to 9:   | 2           |
| 10 to 14: | 4           |
| Total:    | $2 + 4 = 6$ |

In April, **6 cars** with an estimated gas mileage of less than 15 mpg were sold.



Use the histogram above for problems 7 and 8.

- Did Hugo's Used Cars sell more cars with an estimated mileage greater than 25 mpg or less than 25 mpg?
- How many cars did Hugo's Used Cars sell in April?

# Histograms: Examples

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Use this histogram for problems 9 and 10.

9. A grade below 70% is a failing grade. How many students failed the final?

10. **Explain** Students who scored above 70% passed the final. To the nearest percent, what percent of Ms. Jung's class passed the final? Explain your answer.

